

GenAI Guardrail Factory

The safety release gate that lets enterprises ship GenAI with proof, not hope. The missing safety layer for every enterprise GenAI release.

A CI/CD-style gate that adversarially tests any GenAI app, blocks unsafe releases, auto-remediates, and approves only after measurable recovery — turning “it worked in the demo” into audit-ready evidence.

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WHY IT MATTERS

Enterprises are shipping GenAI faster than they can prove it's safe. Validation today is manual — sample prompts, expert review, ad-hoc checks — which can't scale and leaves no audit trail. One hallucination, leaked identifier, or jailbroken approval can become a compliance incident. **Guardrail Factory wraps any GenAI app as a release gate that decides one thing: is this safe enough to ship?** A Google ADK agent runs adversarial tests, scores groundedness, toxicity, and PII against the app's own documents, and a deterministic gate returns **block or approve**. On failure it diagnoses, hardens the prompt, and re-tests — approving only after measurable recovery. Same inputs, same verdict, full evidence on record.

KEY USE CASES

- **One reusable safety gate** — a single standard across BFSI assistants, compliance copilots, support bots, and RAG workflows.
- **Pre-release safety gating** — automated block/approve verdict for any GenAI app before it ships.
- **Regression safety testing** — re-validate after every prompt, model, or document change.
- **Auto-remediation** — diagnose failures and harden prompts, then re-test to recovery.
- **Governance & audit evidence** — store scores, thresholds, verdicts, and timestamps per run.

TARGET PERSONAS

- **AI deployment teams** — a pass/fail gate in the release pipeline.
- **Compliance & risk leaders** — repeatable, evidence-backed sign-off.
- **CISOs & security teams** — PII, jailbreak, and policy-boundary defense.
- **Business / product owners** — confidence that a release is safe to ship.

PROOF & BUSINESS IMPACT

Replaces a 6–18 month manual safety effort with an automated gate that re-runs on every prompt, model, or document change. **In the documented proof-of-concept, a release that should have been stopped was caught, remediated, and recovered — measurably:**

8 → 0	0.84 → 0.94	0.99	0
active failures cleared (PoC run)	groundedness after hardening	PII protection in final run	PII leaks in final run

Bottom line: trust becomes provable before production — cutting breach and compliance exposure (avg. cost of a data breach in India, IBM 2025: ₹22Cr) and turning red-team toil into a repeatable, auditable decision. For TCS, a reusable **GenAI Safety-as-a-Service accelerator** across every regulated client deployment.

GOOGLE PRODUCTS & TECHNOLOGIES USED

Google ADK (LlmAgent) orchestrates the agentic evaluation · **Gemini 2.5 Flash** powers test generation, failure diagnosis, and prompt hardening · **Vertex AI** evaluation patterns · **Cloud Run** hosts the deployed service · **BigQuery** + **Looker** provide the production-scale governance path. Supporting: FastAPI orchestrator, ChromaDB vector store for RAG grounding, SQLite RunStore for PoC run history.

*Model Armor protects the live road. Guardrail Factory decides whether the vehicle is safe enough to enter it. **Good GenAI demos show capability — Guardrail Factory proves deployability.***